

TOPIC 10/60

Tokens and Tokenizers

AI doesn't read letters or words.

It reads Tokens.

Ashish Jangra

linkedin.com/in/ashish-jangra



THE CORE PROBLEM

Neural Networks only understand numbers, not text. How do we feed the word "Hello" into a math equation?

Bad Idea 1: Character Level

A=1, B=2. Too slow, and individual letters don't carry any meaning.

Bad Idea 2: Word Level

Apple=1. The dictionary would be infinite because of typos, slang, and combined words.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE SOLUTION



Subword Tokenization

Instead of letters or whole words, we break language down into common **chunks** (Subwords).

Hamburger becomes:

Ham

+

bur

+

ger

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE MACHINE

The Tokenizer

The Tokenizer is a pre-processing machine. It slices your prompt into tokens **before** it ever reaches the LLM.

The Dictionary

It has a fixed dictionary of about 50,000 to 100,000 specific chunks. Every token is assigned a unique ID number.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



VISUAL EXAMPLE

How it looks

Common words might be a single token. Complex words get chopped up.

"Unbelievable" =

Un believ able



THE ADVANTAGE

Handling the Unknown

Because it uses chunks, the AI can handle words it has never seen before, including typos or brand new slang.

If you type "Supercalifragilistic", the AI doesn't crash because the word isn't in its dictionary.

It just breaks it into familiar chunks: [Super] [cali] [frag] [il] [istic].

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE BLIND SPOT

WHY AI FAILS AT SPELLING

"How many R's are in Strawberry?"

LLMs notoriously fail this. Why? Because they literally cannot "see" the letters.

THE TOKEN VIEW

It sees: [Straw] [berry]

To the AI, these are just two opaque blocks. It doesn't know the block [berry] is spelled with two R's.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE BUSINESS SIDE

The Currency

Tokens are the universal currency of AI. When you use OpenAI or Anthropic APIs, you don't pay per word; you pay per 1,000 tokens.

Rule of Thumb:

1 Token $\approx$ 0.75 English Words.

100 Tokens $\approx$ 75 Words.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



CONTEXT WINDOWS

Memory is Measured in Tokens

When you hear "GPT-4 has a 128K context window", it means it can remember roughly 128,000 tokens in a single chat.

Foreign Languages Cost More

Because tokenizers are optimized for English, a word in Hindi or Japanese might get chopped into 5 or 6 tokens, making it more expensive to process!

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



Tokens & Tokenizers

CONCEPT UNLOCKED



Tokens = Word Chunks



Tokenizer = The Slicer



Explains AI spelling errors



The pricing currency of GenAI

CONNECT WITH ME



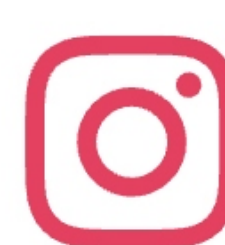
LinkedIn



GitHub



Kaggle



Instagram

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)

